Tropical Cyclone Intensity Estimation From Geostationary Satellite Imagery Using Deep Convolutional Neural Networks

Chong Wang, Gang Zheng, *Senior Member, IEEE*, Xiaofeng Li, *Fellow, IEEE*, Qing Xu, *Member, IEEE*, Bin Liu, *Member, IEEE*, and Jun Zhang, *Senior Member, IEEE*

***Abstract*— In this study, a set of deep convolutional neural networks (CNNs) was designed for estimating the intensity of tropical cyclones (TCs) over the Northwest Pacific Ocean from the brightness temperature data observed by the Advanced Himawari Imager onboard the Himawari-8 geostationary satellite. We used 97 TC cases from 2015 to 2018 to train the CNN models. Several models with different inputs and parameters are designed. A comparative study showed that the selection of different infrared (IR) channels has a significant impact on the performance of the TC intensity estimate from the CNN models. Compared with the ground truth Best Track data of the maximum sustained wind speed, with a combination of four channels of data as input, the best multicategory CNN classification model has generated a fairly good accuracy (84.8%) and low root mean square error (RMSE, 5.24 m/s) and mean bias (−2.15 m/s) in TC intensity estimation. Adding attention layers after the input layer in the CNN helps to improve the model accuracy. The model is quite stable even with the influence of image noise. To reduce the side-effect of the very unbalanced distribution of TC category samples, we introduced a focal_loss function into the CNN model. After we transformed the multiclassification problem into a binary classification problem, the accuracy increased to 88.9%, and the RMSE and the mean bias are significantly reduced to 4.62 and −0.76 m/s, respectively. The results show that our CNN models are robust in estimating TC intensity from geostationary satellite images.**

***Index Terms*— Convolutional neural network (CNN), deep learning, remote sensing, tropical cyclone (TC).**

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Chong Wang is with the Key Laboratory of Marine Hazards Forecasting, Ministry of Natural Resources, Hohai University, Nanjing 210098, China.

Gang Zheng is with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou 310012, China, and also with the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai 519082, China.

Xiaofeng Li is with the CAS Key Laboratory of Ocean Circulation and Waves, Institute of Oceanography, Chinese Academy of Sciences, Qingdao 266071, China, and also with the Center for Ocean Mega-Science, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: xiaofeng.li@ieee.org).

Qing Xu was with the Key Laboratory of Marine Hazards Forecasting, Ministry of Natural Resources, Hohai University, Nanjing 210098, China. She is now with the College of Information Science and Engineering, Ocean University of China, Qingdao 266100, China (e-mail: maggiexu@hhu.edu.cn).

Bin Liu is with the School of Marine Sciences, Shanghai Ocean University, Shanghai 201306, China.

Jun Zhang is with the Hurricane Research Division, NOAA/AOML, University of Miami/CIMAS, Miami, FL 33149 USA, and also with the University of Miami/CIMAS, Miami, FL 33149 USA.

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I. Introduction

TROPICAL cyclones (TCs) can cause enormous damage to human lives and properties owing to the destructive winds, severe flooding, and coastal inundation from storm surges. The accurate estimation of a TC intensity is essential for forecasters and emergency responders. Different types of satellite remote sensing technologies have provided many practical methods for TC monitoring [1]–[8]. Operational forecast centers apply a variety of data sets to estimate TC intensity. Among the most frequently applied data sets are intensity estimates derived from the traditional, subjective Dvorak technique [9] and automated Dvorak algorithms [10], both of which rely on infrared (IR) satellite imagery. Other algorithms used to derive TC intensity from IR satellite data include the deviation-angle variance technique (DAVT) [5], [13], [14] and the convolutional neural network (CNN) [15]–[19], which are still in scientific research. Additional widely used methods to estimate TC intensity from satellite data include microwave sounder-based methods [20]–[22] and consensus methods [23], among others.

The classic Dvorak method [9] relates a TC rotation, eye shape, and deep thunderstorms to its enhancement or weakening. The technique assumes that cyclones with similar intensities tend to have similar patterns. It requires an expert to visually analyze a TC cloud structure in visible and IR images. Once a pattern is detected, features such as cloud organization properties, inner core characteristics, curvature radius, and rain rate are further analyzed to estimate the intensity [11], [12]. Although widely used, the Dvorak technique is subject to human errors and has some inherent limitations. The latest advanced Dvorak technique [10] applies automated algorithms to remove human subjectivity.

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The DAVT technique is based on a directional gradient statistical analysis of IR satellite imagery to estimate the TC intensity over the North Atlantic [13] and North Pacific [14]. A priori cyclone center information is necessary for this approach. Although the DAVT technique can automatically locate the TC center, the location error may lead to slight uncertainty in intensity estimation.

TC intensity estimation can also be performed using the CNN method [24]. CNN is a feedforward deep neural network inspired by the biological natural visual cognition mechanism [25]. This algorithm avoids the complicated preprocessing of images and has achieved great success in image classification, target recognition, and other research fields [25]–[27]. Recently, several CNN models have been developed to estimate TC intensity from IR satellite imagery by using the direct classification method [18], [19] or the indirect regression method [15]–[19]. The classification method first classifies a cyclone according to the intensity category (see Table I) and then calculates the maximum wind speed (MWS) of TCs. Compared with the Dvorak or DAVT technique, the CNN model obtained a higher accuracy [15]–[19].

The CNN model has produced a relatively high accuracy in estimating TC intensity for some ocean regions [15], [16], [18], [19]. Nevertheless, there are still some key issues to be solved. First, for satellite sensors with several IR channels, each channel provides a different distribution of brightness temperature. These data reveal the spectral characteristics of various atmospheric and oceanic variables, and some of them are closely linked with the TC structure, such as the cloud top, the lower/middle layer cloud, the water vapor, and sea surface temperature (SST). It has been pointed out that the cyclone intensity is proportional to the difference between the absolute temperature of the boundary layer and that of the storm top [28], and related to the upper tropospheric brightness temperature anomalies [29] and the water vapor in the upper and middle layers [16]. Recently, Sieron *et al.* [30] found that the intensity of a cyclone is necessarily directly related to the temperature deficit of cloud top versus sea surface, and the surplus in saturation entropy in the eyewall versus its surroundings. Therefore, the data from multiple channels can capture the atmospheric and sea surface variables that are closely related to TC intensity better than data from a single or two channels can. The accuracy of the CNN model in estimating TC intensity would be improved if multichannels of satellite data are utilized as input. However, only one or two channels of data have been used in previous studies. It is necessary to investigate the contribution of multichannel data and find an optimal channel combination for tropical intensity estimation.

It is also a great challenge to develop a robust CNN with a small number of samples and an unbalanced distribution of samples of different TC categories. To avoid the side-effect of an unbalanced data set, Pradhan *et al.* [18] tried to balance the distribution at the cost of reducing training data size. However, this approach cannot solve the data imbalance problem fundamentally.

In this study, we developed a group of CNN models and selected the best one to estimate the TC intensity over the Northwest Pacific using high temporal and spatial coverage of Himawari-8 images. Our goal is to develop an objective and efficient TC intensity estimation method. We organize this article as follows. Section II introduces the data set and data pre-processing. The CNN models and experimental schemes are presented in Section III. The performance analysis and discussion are given in Section IV. The conclusion is given in Section V.

II. Data Set

A. Best Track Data Set of TCs

The Best Track data set for TCs provided by the Shanghai Typhoon Institute, China Meteorological Administration (STI/CMA), Beijing, China [32] was used to label Himawari-8 (H-8) images. The track and MWS of a TC are recorded every 3 or 6 h in this data set. The 6-h Best Track data were all interpolated to 3 h to increase the data samples for TC intensity estimation.

B. Geostationary Satellite Imagery of TCs

H-8 is a geostationary meteorological satellite launched by the Japan Meteorological Agency (JMA), Tokyo, Japan, in October 2014. The Advanced Himawari Imager (AHI) onboard H-8 provides 10-min full-disk coverage in the Northwest Pacific from 16 channels: three visible, three near-IR, and ten thermal IR channels. The time interval of full-disk observations is 10 min. We downloaded the brightness temperature data from five IR channels (Channels 7, 8, 13, 14, and 15, see Table IV for details) from http://www.eorc.jaxa.jp/ptree. These five channels are for the monitoring of low-level clouds, natural disasters (Channel 7), water vapor in the upper and middle layers (Channel 8), and cloud top and SSTs (Channels 13–15) [31]. We obtained 6.690 five-channel composite images with a 3-h time interval and a 5-km resolution, which capture 97 TCs over the Northwest Pacific between 2015 and 2018. For each composite image, we extracted a 251×251 pixel subimage covering an area of 1255 km $\times$ 1255 km centered on the TC center determined from the Best Track data set of TCs, as shown in Fig. 1. One can see that each channel provides unique information about a TC. These 3-h composite images were labeled carefully according to eight categories using the Saffir–Simpson TC wind scale ($H1$–$H5$) along with intensity categorization for the tropical storm (TS) and tropical depression (TD) as TC intensity categories (Table I). The images were randomly divided into training, validation, and test data at a ratio of 3:1:1, i.e., 4.014 images for training, 1.338 for validation, and 1.338 for testing.

C. Data Preprocessing

Data normalization before the neural network training is required to speed up the calculations and obtain good results [33]. In this study, satellite-observed brightness temperature data were linearly transformed to the interval [0, 1] by

$$y = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of the brightness temperature x from all the five channels

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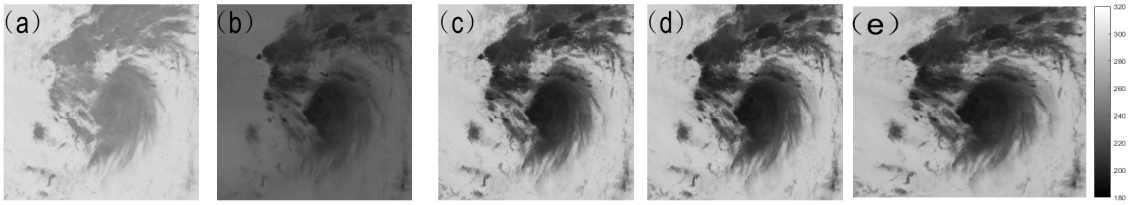


Fig. 1. Brightness temperature (K) images of TC Halola from different channels. (a) Channel 7, (b) Channel 8, (c) Channel 13, (d) Channel 14, and (e) Channel 15, with a spatial coverage of 1255 km × 1255 km. The images were acquired at 0:00 UTC on July 13, 2015.

TABLE I
SAFFIR–SIMPSON HURRICANE WIND SCALE AND RELATED CLASSIFICATIONS [18]

Category	Symbol	Wind speed (m/s)
No Category	NC	≤ 10.3
Tropical depression	TD	10.3 - 17.0
Tropical storm	TS	17.1 - 32.4
Category 1	H1	32.5 - 42.2
Category 2	H2	42.3 - 48.9
Category 3	H3	49.0 - 57.6
Category 4	H4	57.7 - 70.0
Category 5	H5	≥ 70.0

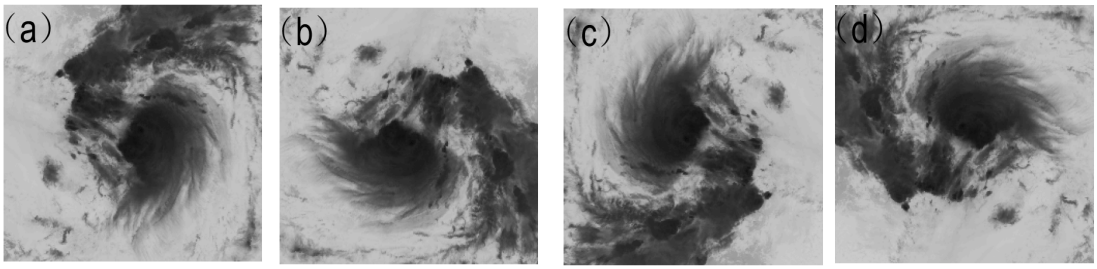


Fig. 2. Transformation of Fig. 1(e). (a) Original image. (b) 90° rotation. (c) 180° rotation. (d) 270° rotation.

(7, 8, 13, 14, and 15); y is the normalized pixel value limited to [0, 1].

Studies show that the rotation of images in a CNN model can reduce the orientation sensitivity and does not affect the classification accuracy [25], [34]. Therefore, the normalized training and validation images were artificially rotated 90°, 180°, and 270° clockwise, as shown in Fig. 2. In this way, the number of samples increased by three times, and we obtained 16.056 training and 5.352 validation images (Table II).

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TEST DATA FOR THE CNN-BASED TC INTENSITY ESTIMATION MODEL

Category	Training	Validation	Test	Total
NC	402	134	35	571
TD	4773	1591	386	6750
TS	6474	2158	552	9184
H1	2226	742	178	3146
H2	885	295	85	1265
H3	747	249	62	1058
H4	498	166	36	700
H5	51	17	4	72
Total	**16056**	**5352**	**1338**	**22746**

III. DEEP CONVOLUTIONAL NEURAL NETWORKS FOR TC INTENSITY ESTIMATION

A. CNN Model Configuration

A simple CNN model contains various combinations of convolutional, pooling, and fully connected (FC) layers. The convolutional layer extracts image features. The pooling layer then filters the maximum value of these features to reduce the number of features. The FC layer learns all the features and eventually output the results. To prevent over-fitting, a dropout layer [35], which means setting the partial weight to zero, is usually added before the FC layer. A deep CNN is generally designed as a feedforward network and can be trained with the backpropagation algorithm. As the errors backpropagate in a network, CNN is optimized by updating the weights and biases to minimize the loss function [25].

An example architecture of our CNN models is shown in Fig. 3. It consists of one input layer, four convolutional layers, four pooling layers, two FC layers, and one output layer. In this study, various network configuration performances in estimating the location and intensity of TCs were investigated. Table III lists the parameter sets in one example of the CNN ($M1$–$M12$ in Table IV) models. The parameters include the input shape, filter shape, output size, the number of parameters, along with the number of kernels used in each layer, and corresponding stride and padding. The output images for each layer are shown in Fig. 6. With the configuration of one

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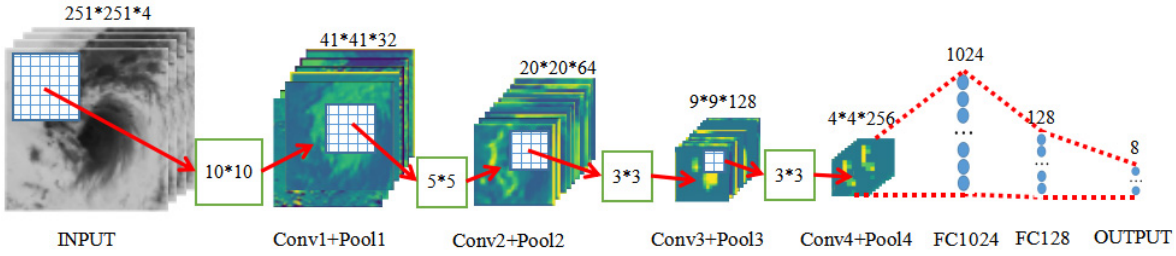


Fig. 3. Architecture of one of our CNN models ($M1$–$M12$ in Table IV).

TABLE III

CONFIGURATION OF A CNN MODEL ($M12$ IN TABLE IV) USED IN THIS STUDY. "s" IS THE STRIDE, AND "p" IS THE NUMBER OF ZERO-PADDING. FOR EXAMPLE, CONV1(32 AT 3×3, $s = 2$, AND $p = 0$) DENOTES 32 KERNELS IN THE FIRST CONVOLUTIONAL LAYER WITH A SIZE OF 3×3, AND THE STRIDE AND ZERO-PADDING ARE 2 AND 0, RESPECTIVELY

Layer	Activation function	Shape	Output size	Number of parameters
Input		4@251×251		
Conv1	Relu	16@10×10, s=3, p=0	84×84	12832
Pool1		3×3, s=2, p=0	41×41	0
Conv2	Relu	32@5×5, s=1, p=2	41×41	51264
Pool2		3×3, s=2, p=0	20×20	0
Conv3	Relu	64@×3×3, s=1, p=2	20×20	73856
Pool3		3×3, s=2, p=0	9×9	0
Conv4	Relu	128@×3×3, s=1, p=2	9×9	295168
Pool4		3×3, s=2, p=0	4×4	0
Dropout (0.5)		4096	4096	0
Fc1	Relu	1024	1024	4195328
Dropout (0.5)		1024	1024	0
Fc2	Relu	128	128	131200
Dropout (0.5)		128	128	0
Output	softmax	8		8200
Total				4759824

TABLE IV

TC INTENSITY CLASSIFICATION RESULTS USING H-8 IMAGES FROM DIFFERENT CHANNELS

Model ID	H-8 Image Channel (wavelength [μm])	Accuracy	RMSE (m/s)
M1	7 (3.9)	78.1%	6.24
M2	8 (6.2)	78.2%	6.38
M3	13 (10.4)	80.3%	5.86
M4	14 (11.2)	80.3%	5.93
M5	15 (12.3)	80.5%	5.80
M6	7,15	81.0%	5.78
M7	8,15	81.3%	5.67
M8	13,15	81.5%	5.68
M9	14,15	80.7%	6.02
M10	7,13,15	82.7%	5.53
M11	8,13,15	82.5%	5.60
M12	7,8,13,15	82.9%	5.47

CNN model in Table III, four-channel input data of 251×251 size are convolved in the first convolutional layer (Conv1, 16 kernels of 10×10 size, with stride $= 3$, and padding $= 0$). In this process, 16 feature maps of 84×84 size can be extracted, and 12832 parameters are learned. The first pooling layer (Pool1) then produces maps of 41×41 size by filtering these feature maps. In the dropout layer [35], half of the 128 feature maps generated with Conv3 move to the next layer.

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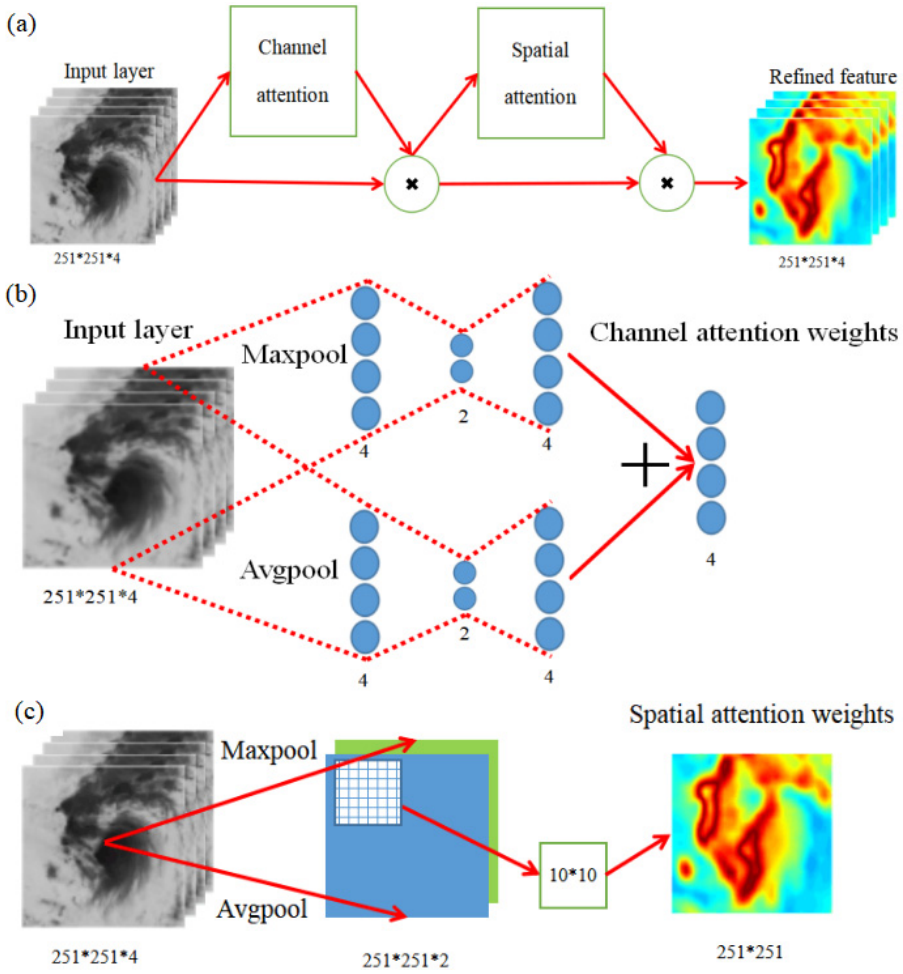


Fig. 4. Diagram of the attention layers in the CNN model. (a) Overview of the attention layers. (b) Diagram of the channel attention layer. (c) Diagram of the spatial attention layer. "+" and "×" are plus and multiplication signs, respectively.

We obtain eight output values from the CNN model (hereafter called the multicategory CNN classification model). The nth value represents the confidence with which a TC is classified as the nth intensity category.

In addition, the channel attention layer and the spatial attention layer are also added to our model between the input layer and the first convolutional layer (Fig. 4). By generating the weights of different channels, the channel attention layer can tell us which channels are more important in the CNN model. Similarly, the spatial attention layer will learn the most relevant region for intensity estimation [36]. As a result, satellite images of TCs at different categories will be treated differently in the CNN model, as shown in Fig. 5 in Section IV-B.

B. Setup of Comparative Experiments

The input satellite imagery quality and the parameter setting of a network affect the performance of a CNN model. To achieve an optimal TC monitoring scheme in this study, we designed numerous CNN model experiments with various configurations to locate a TC and estimate its intensity from satellite images with different channel combinations.

To address the question of which channel works better, we set up 12 experiments ($M1$–$M12$ in Table IV) to evaluate the CNN model's performance using H-8 images from five IR channels. With the basic CNN configuration shown in Table III, these CNN models' accuracies were first analyzed. Here, we adopted the nonlinear Relu as the activation function in the inner product calculation of convolutional and FC layers. The optimization function is Adam [37]. With an optimal channel combination of satellite imagery as input, CNN models' performances with different configurations (Table V) were then investigated.

IV. Results and Discussion

Similar to the criteria in [18], we evaluate our CNN models' performance concerning several aspects.

1) *Accuracy:* Accuracy measure is the number of exact-hits, which corresponds to the correct classification with the highest confidence.
2) *Root Mean Square Error of TC Intensity:* The estimated intensity or MWS (W) of a TC is defined as the weighted average of the representative wind speeds associated with the two highest probability categories

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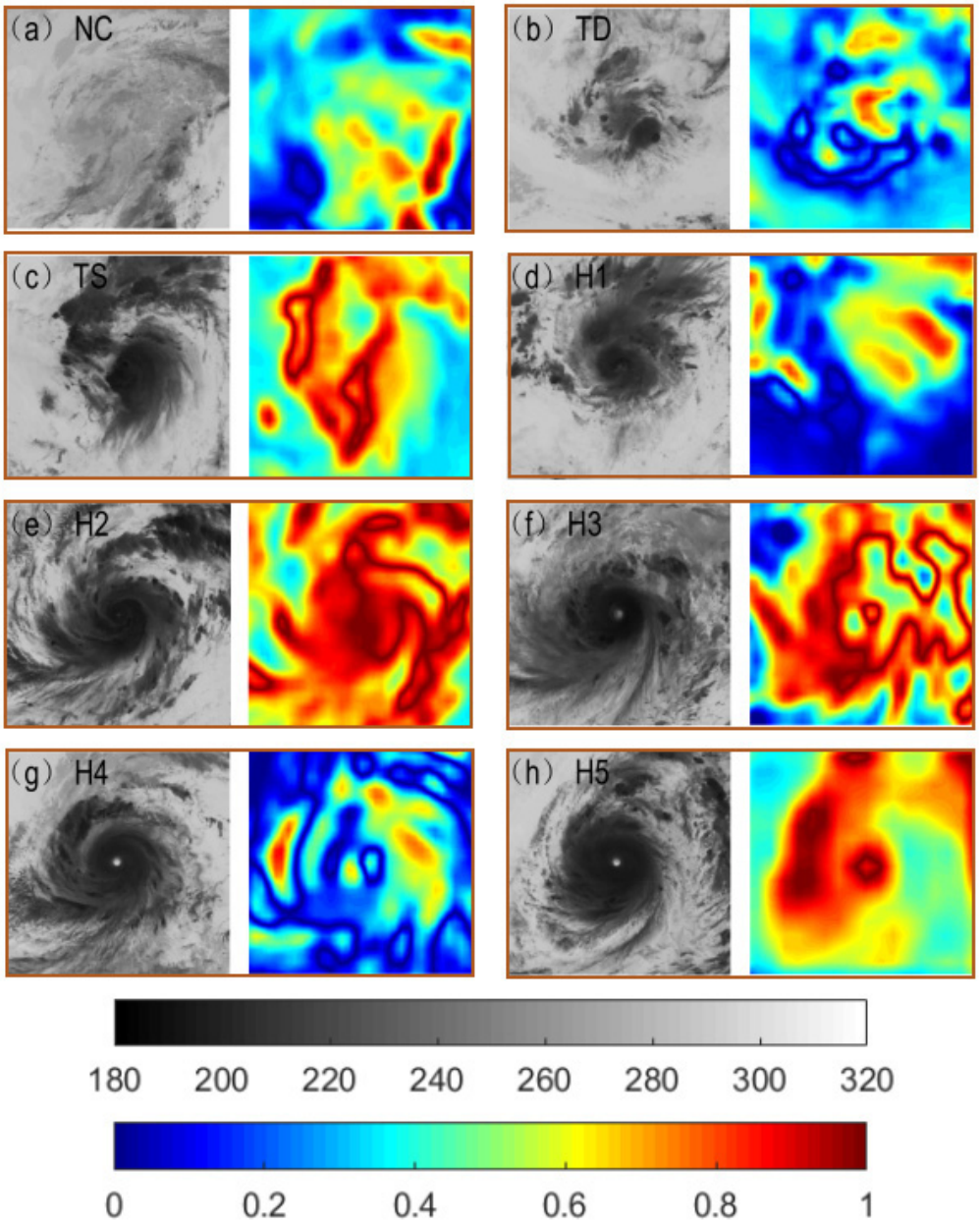


Fig. 5. Original H-8 image [(Left) unit: Brightness temperature, K] and (Right) heat map of the spatial attention layer in the CNN model M16. (a)–(h) TC with category of NC–$H5$, respectively.

which can be evaluated by [18]

$$W = U_1 \times P_1 + U_2 \times P_2 \tag{2}$$

where P_1 and P_2 are the model outputs of the confidence of the TC categories with the highest and second-highest confidence, respectively; U_1 and U_2 are the representative wind speeds of the corresponding category. If the output of the model is [NC $= 0.0$, TD $= 0.0$, TS $= 0.8$, $H1 = 0.2$, $H2 = 0.0$, $H3 = 0.0$, $H4 = 0.0$, $H5 = 0.0$], then $P_1 = 0.8$, $P_2 = 0.2$, and $W = U_{\mathrm{TS}} \times 0.8 + U_{H1} \times 0.2$.

Table I shows the wind speed range for different TC categories from TD to $H4$. The representative wind speed is the median value of the speed range. For categories NC and $H5$, which are not given the clear speed range in Table I, the representative wind speeds are calculated by averaging all the MWSs within each category. For example, we have collected 21 $H5$ category images in this study, and the representative wind speed of these 21 $H5$ storms is 76.0 m/s.

3) *Confusion Matrix and Classification Report*: As shown in Table VI, the confusion matrix shows the overall classification performance of a model. In a confusion matrix, the number along the diagonal line represents the number of correctly classified images for any category. The classification report (Table VII) describes the precision (P), recall (R, the confidence of detection), and f1-score ($F1$) of a CNN model. P and R are the ratios of the number of actual positive class values to the total number of positive classifications and the

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TABLE V

TC INTENSITY CLASSIFICATION RESULTS FROM DIFFERENT NETWORK CONFIGURATIONS USING FOUR-CHANNEL (CHANNELS 7, 8, 13, AND 15) COMPOSITE H-8 DATA. TAKING THE CONFIGURATION OF $M14$ AS AN EXAMPLE, THERE ARE FOUR CONVOLUTIONAL LAYERS (C1–C4) AND FOUR POOLING LAYERS (P1–P4), C32 AT 10×10 DENOTES 32 KERNELS IN THE FIRST CONVOLUTIONAL LAYER WITH A SIZE OF 10×10, AND THE STRIDE AND ZERO-PADDING ARE 3 AND 0, RESPECTIVELY

Model ID	Configurations	Accuracy	RMSE (m/s)
M12	C1(16@10×10,s=3,p=0), P1(3×3,s=2,p=0), C2(32@5×5,s=1,p=2), P2(3×3,s=2,p=0), C3(64@3×3,s=1,p=2), P3(3×3,s=2,p=2), C4(128@3×3,s=1,p=2), P4(3×3,s=2,p=2), Dropout=0.5, FC1024, FC128	82.9%	5.47
M13	C1(32@10×10,s=3,p=0), P1(3×3,s=2,p=0), C2(64@5×5,s=1,p=2), P2(3×3,s=2,p=0), C3(128@3×3,s=1,p=2), P3(3×3,s=2,p=2), C4(256@3×3,s=1,p=2), P4(3×3,s=2,p=2), Dropout=0.5, FC1024, FC128	84.8%	5.24
M14	C1(64@10×10,s=3,p=0), P1(3×3,s=2,p=0), C2(128@5×5,s=2,p=2), P2(3×3,s=2,p=0), C3(256@3×3,s=1,p=2), P3(3×3,s=2,p=2), C4(512@3×3,s=1,p=2), P4(3×3,s=2,p=2), Dropout=0.5, FC1024, FC128	80.1%	5.90
M15	C1(32@10×10,s=3,p=0), P1(3×3,s=2,p=0), C2(64@5×5,s=2,p=2), P2(3×3,s=2,p=0), C3(128@4×4,s=2,p=2), P3(3×3,s=2,p=2), C4(256@3×3,s=2,p=2), P4(3×3,s=2,p=2), Dropout=0.5, FC1024, FC128	77.8%	6.47
M16	Channel Attention Layer Spatial attention layer C1(32@10×10,s=3,p=0), P1(3×3,s=2,p=0), C2(64@5×5,s=1,p=2), P2(3×3,s=2,p=0), C3(128@3×3,s=1,p=2), P3(3×3,s=2,p=2), C4(256@3×3,s=1,p=2), P4(3×3,s=2,p=2), Dropout=0.5, FC1024，FC128	86.0%	5.17

TABLE VI

CONFUSION MATRIX USING THE MULTICATEGORY CNN CLASSIFICATION MODEL $M16$

		MODEL CATEGORY								
		NC	TD	TS	H1	H2	H3	H4	H5	Total
ACTUAL CATEGORY	NC	22	11	2	0	0	0	0	0	35
	TD	3	349	29	4	0	1	0	0	386
	TS	0	29	495	27	0	1	0	0	552
	H1	0	1	18	145	12	2	0	0	178
	H2	0	0	3	14	58	8	2	0	85
	H3	0	0	1	2	6	48	5	0	62
	H4	0	0	1	2	0	4	29	0	36
	H5	0	0	0	0	0	0	1	3	4
	Total	25	390	549	194	76	64	37	3	1338

TABLE VII

CLASSIFICATION REPORT OF THE MULTICATEGORY CNN CLASSIFICATION MODEL $M16$

Category	P	R	F1	Total
NC	0.63	0.88	0.73	35
TD	0.90	0.89	0.89	386
TS	0.89	0.90	0.89	552
H1	0.81	0.74	0.78	178
H2	0.68	0.76	0.72	85
H3	0.77	0.75	0.76	62
H4	0.80	0.78	0.79	36
H5	0.75	1.00	0.85	4
Avg.	0.86	0.86	0.86	1338

number of positive class values in the test data. $F1 = 2P \times R/(P + R)$ is the harmonic mean of recall and precision. The closer the three values are, the more stable the model is.

A. Model Performance

In this section, the CNN model's accuracy with different input data or network configurations is analyzed by comparing the model result with the Best Track data set.

Table IV shows the TC classification results for the CNN models using the 1338 testing images from different channels or channel combinations. The network configuration listed in Table III was used. For single-channel input to CNN models ($M1$–$M5$), we found that the highest accuracy was obtained with either Channel 14 or 15 as input. Both channels are in the thermal-IR bands, often used for cloud detection and SST calculation. The result confirms the conclusion of Sieron *et al.* [30] that the TC intensity is correlated with the temperature deficit of cloud top versus sea surface. As shown

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in Table IV, model $M5$ produces the best result with the smallest root mean square error (RMSE) in intensity estimation of the five single-channel input models. For testing using combinations of multiple channels, we first combined Channel 15 with the other channels, respectively, for $M6$–$M9$. Comparing $M9$ with $M5$, we can see that Channel 14 makes little contribution to the improvement of the model because its wavelength is close to Channel 15. These two thermal IR channels provide duplicate information for TC intensity estimation. CNN models $M6$–$M8$ show improvement when adding other IR channels, with $M8$ having a slightly better advantage.

We then investigated the influence of using three-channel or four-channel data on the accuracy of our CNN models. Among different combinations, inputting four-channel data into $M12$ generates the best result in Table IV. The accuracy reaches up to 82.9%, suggesting that this CNN model based on inputs from four-channel IR images delivers results that are more consistent with observations. Consistent with the previous theoretical study [28]–[30], our results indicate that the additional information of the water vapor or cloud characteristics provided by multichannels does play a significant role in the estimation of TC intensity. With multichannel images as input, CNN can learn more complex features of this complicated weather process, which may have been overlooked by the traditional methods.

Once we have found the best channel combination as input into the CNN model, $M12$, a variety of network configurations were further tested. The results are listed in Table V. CNN models $M12$–$M15$ consist of four convolutional layers, four pooling layers, and two FC layers. For $M12$–$M15$, each model holds a different number of kernels in the convolutional layers. In general, the model's accuracy is slightly increased with the increase in the number of kernels ($M12$ and $M13$) but then significantly reduced ($M12$) beyond a specific range of the kernel number. Although more feature maps can be extracted with more kernels, these maps may not always positively impact on the improvement of the model. This degradation may also be related to the relatively fewer training data in our model. The number of parameters to be learned by the model is more significant than that of the training data, making it challenging to generate good results. Compared with $M14$, the accuracy of $M15$ is even lower despite using different strides. If we add one channel attention layer and one spatial attention layer behind the input layer of $M14$ in $M16$, the CNN model's performance is further improved. $M16$ has generated a fairly good accuracy (86.0%) and low root mean square error (RMSE, 5.17 m/s). This improvement indicates that the attention layers can help the model to focus on the key factors.

B. Discussion

1) Features Learned by CNN: In the process of training the CNN models, the neural network learns many texture features of TCs. The correlation between these features and TC intensity determines the accuracy of the model. Using a deep visualization toolbox [38], we can analyze the image features learned by each network layer. We take the configuration of the CNN model $M16$ as an example to explain the features.

With four-channel H-8 images as input in $M16$, which involves channel and spatial attention layers, four weights were generated from the channel attention layer, with values of 0.3, −0.27, −1, and 0.75 for Channels 7, 8, 13, and 15. This implies that Channels 13 and 15 play a significant role in the CNN model M16, while Channels 7 and 8 play a supporting role, which is in good agreement with the results in Table V.

For TCs at different categories or stages, the CNN model focuses on different regions by utilizing the spatial attentional layer. Fig. 5 shows the heat map, and the value "1" represents the region of most interest in model $M16$. One can see the model focuses on the area with low brightness temperature for most category TCs except $H4$. For stronger TCs, $H2$, $H3$, and $H5$, these regions display clearly the eye and shape of the TC.

There are 32 feature maps generated from the first convolutional layer in $M16$, 16 of which are shown in Fig. 6(a). One can see the overall structure of a TC with the prominent eye located at the center of the image. The first pooling layer reduces the number of the extracted features through the maximum filter and makes them more blurred. The second convolutional layer extracts more image features, some of which have become abstract [Fig. 6(b)]. By further studying these features in deeper layers, we got images with more abstract characteristics [Fig. 6(c) and (d)]. All these feature maps demonstrate the central dense overcast and structure of a TC. The multichannel data discovered these features, and some of them might show the TC-related characteristics revealed in previous studies.

Traditional techniques are mostly based on human understanding of TCs and manual analysis of specific features (e.g., spiraling, central cold clouds, and asymmetries). TC intensity is a function of these features. However, traditional techniques ignore features that have not yet been discovered but may be more relevant to TCs. Fundamentally, these methods are still subjective. CNN can extract more features from TC images, which may be undiscovered, and establish a relationship between these features and TC intensity. The CNN method is entirely objective, although there are no tools to analyze CNN-derived results intuitively. It may be possible to discover some unknown features that affect TCs by combining the information or features extracted from the CNN with the increasing understanding of physical mechanisms of TCs.

2) Unbalanced Distribution of TC Categories: In general, the above multicategory CNN classification model $M16$ performs reasonably well in estimating TC intensity. However, as Tables VI and VII show the classification results are not very satisfactory for categories with small training samples. For example, only 58 out of 85 $H2$-level TCs were correctly classified, and the precision is only 68.2%. The degradation is due to the unbalanced distribution of TC category samples while we trained the CNN-I model. Taking H2 as an example, the number of training data in this category only accounts for

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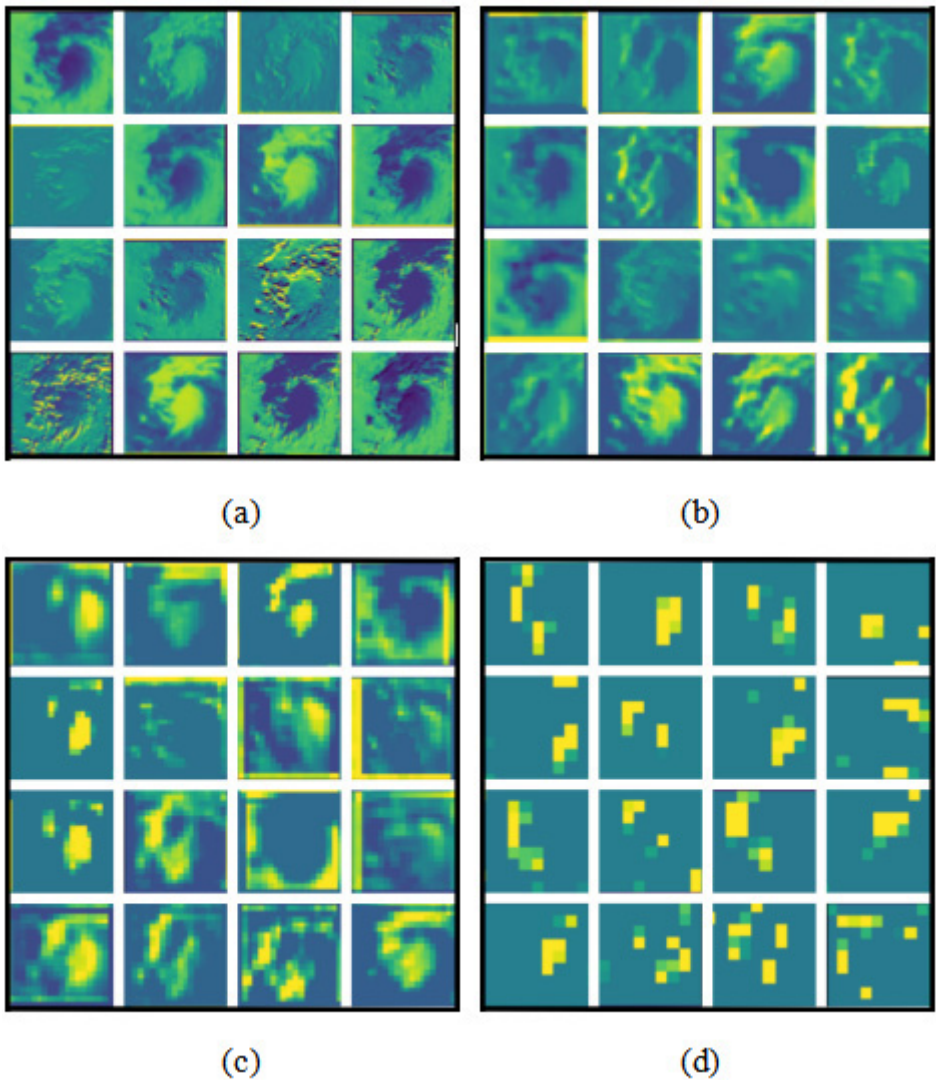


Fig. 6. Feature maps generated from different layers in the CNN model $M16$. (a) First convolutional layer. (b) Second convolutional layer. (c) Third convolutional layer. (d) Last convolutional layer. The input data are four-channel (Channels 7, 8, 13, and 15) H-8 images (Fig. 1) of TC Halola in July 2015.

5.5% of the total training number (Table II). The loss in a CNN model would only increase slightly, even if most $H2$-level cyclones are misclassified. Since CNN updates each layer's weight value according to the loss, the network cannot learn the features of a category well with very few samples.

To reduce the side-effect of unbalanced training data, we can ideally eliminate part of the data to keep the number of TC data in each category balanced [18], [39]. However, this reduction would dramatically reduce the number of training data. Another possible solution is to use a focal_loss (FL) function in a model. This function helps a model learn the features better by increasing the weight of the category with fewer data in the loss. The function has shown its superiority in the field of target recognition [40]. As a result, we changed the loss function of our best CNN model $M16$ to FL to see if it can help to improve the classification accuracy.

The FL is [40]

$$\mathrm{FL}(p_t) = -a_t(1-p_t)^r \log(p_t) \tag{3}$$

where $p_t \in [0, 1]$ is the model output of confidence for a category; $a_t \in [0, 1]$ is the weight coefficient (the smaller the number of data in this category, the higher the value of a_t); and r is the tunable focusing parameter. For each TC category, the values of a_t and r, determined by their proportions in the total data set, are listed in Table VIII. As shown in Table IX, by using the FL function in the multiclassification model $M17$, the TC intensity estimation accuracy is increased to 86.6%, and the RMSE is reduced by 2.1%.

It is well known that a TC at different stages has different related features. One multiclassification model may not capture all these differences well. Hsu pointed out that, in many cases, the multiclassification problem can be transformed into a target recognition or binary-classification problem [41]. If we use eight binary models, with each model learning the TC relevant feature at the corresponding stage, better results may be obtained. Therefore, in model $M18$, eight binary CNN classification models were also set up to identify TC categories from NC to $H5$. The configuration of each model is the

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TABLE VIII
VALUES OF a_t AND r USING THE FL FUNCTION FOR EACH TC CATEGORY

Category	NC	TD	TS	H1	H2	H3	H4	H5
a_t	0.97	0.70	0.60	0.86	0.94	0.95	0.97	0.997
r	3	3	3	3	3	3	3	3

TABLE IX
COMPARISON OF CNN MODELS' PERFORMANCE USING MULTICATEGORY CLASSIFICATION AND BINARY CLASSIFICATION SCHEMES

Model ID	Model scheme	Accuracy	RMSE (m/s)
M16	Multi-category classification	86.0%	5.17
M17	Multi-category classification + *focal_loss*	86.6%	5.06
M18	Binary classification + *focal_loss*	**88.9%**	**4.62**

TABLE X
CONFUSION MATRIX USING THE BINARY CNN CLASSIFICATION MODEL $M18$

		MODEL CATEGORY								
		NC	TD	TS	H1	H2	H3	H4	H5	Total
ACTUAL CATEGORY	NC	24	11	0	0	0	0	0	0	35
	TD	2	354	25	2	2	1	0	0	386
	TS	0	24	511	14	2	1	0	0	552
	H1	0	2	14	152	7	3	1	0	178
	H2	0	0	1	12	62	9	1	0	85
	H3	0	0	0	3	4	52	3	0	62
	H4	0	0	0	0	2	4	30	0	36
	H5	0	0	0	0	0	0	0	4	4
	Total	26	391	551	183	79	70	35	4	1338

same as that of $M16$. The FL with values of a_t and r listed in Table VIII was also used. Classification label "1" or "0" represents a TC category. For example, suppose the MWS of a TC on a satellite image is 27.0 m/s, which corresponds to the TS category. In that case, the image label is "1" in the TS binary classification model and "0" in the training of other binary classification models. In this way, we only need to adjust each binary classification model as accurately as possible.

When a TC image is input to eight CNN models, each model will generate an output with a value ranging from 0 to 1. The confidence of each TC category, for example, the TS category $P_{\mathrm{TS}'}$ is

$$P_{\mathrm{TS}'} = [(1 - P_{\mathrm{NC}}) + (1 - P_{\mathrm{TD}}) + P_{\mathrm{TS}} + (1 - P_{H1}) + (1 - P_{H2}) + (1 - P_{H3}) + (1 - P_{H4}) + (1 - P_{H5})]/8 \quad (4)$$

where P_{NC}–P_{H5} are the outputs of eight CNN binary models.

The corresponding highest confidence number finally determines the intensity category of the TC and the MWS is calculated as

$$W = U_1 \times P_1' + U_2 \times \left(1 - P_2'\right) \quad (5)$$

where P_1' and P_2' are defined as the confidences of the TC categories with the highest and second-highest confidences, respectively; U_1 and U_2 are the representative wind speeds of the corresponding TC categories, respectively.

From Table IX, one can see the binary classification model $M18$ generates the best results. The accuracy is much higher than the multiclassification model $M16$. Compared with $M17$, which also adopts the FL function, the RMSE of model $M18$ is greatly decreased to 4.62 m/s, and the correlation coefficient reaches up to 0.95 (Fig. 7). This improvement suggests that both the introduction of the FL function and the binary CNN classification model contribute to the improvement of the TC intensity estimation results in the case of unbalanced training data.

Tables X and XI show the confusion matrix and classification report of the binary classification model $M18$, which adopts the FL function. Compared with the results from the multiclassification model (Tables VI and VII), the numbers of exact-hits for NC, TD, TS, $H1$, $H2$, $H3$, and $H4$ are all increased. Fewer $H1$ and $H4$ TCs are misclassified as their adjacent categories. The $H1$–$H4$ classification precisions significantly improved by 4.9%, 7.4%, 9.1%, and 3.8%. As shown in Fig. 8, compared with the results of model $M17$, the RMSEs of $M18$ for TD, TS, $H1$, $H2$, $H3$, $H4$, and $H5$ are all decreased. The biases for all categories in $M18$ are closer to 0, while $M17$ tends to produce negative biases.

We also compared our binary CNN classification model's performance in estimating TC intensity with the techniques listed in Table XII. Our model exhibits a smaller RMSE than the DAVT technique and most CNN regression or classification models. A comparative study on TC intensity estimation over different regions carried out by Chen *et al.* [15] showed that the result over the West Pacific was much worse than that over the global ocean. The worse result is attributable to higher

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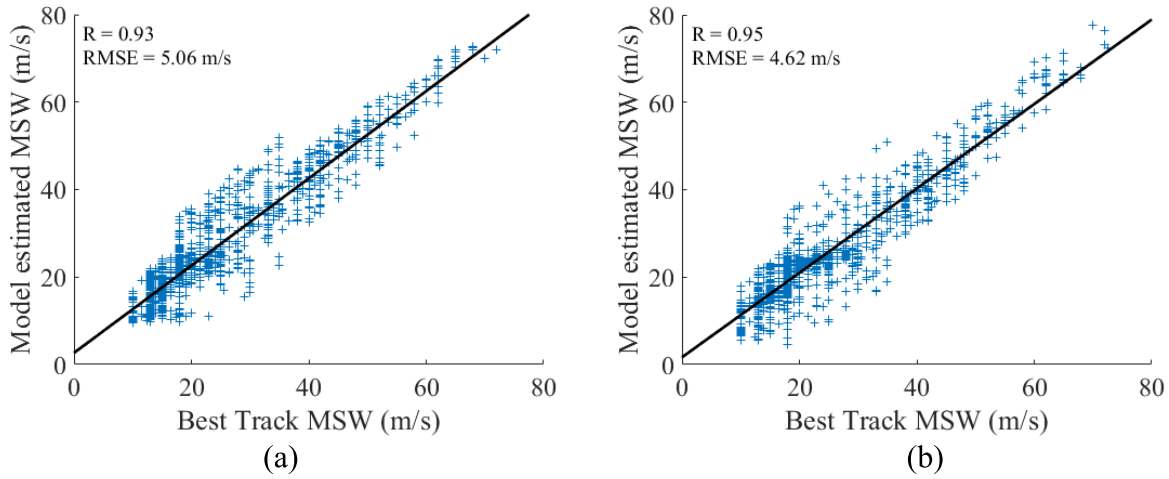


Fig. 7. (a) Comparison of the TC MSW from the Best Track data set with that estimated from model (a) $M17$ and (b) $M18$. The test data set consists of 1338 images. The least-squares regression line (black line), correlation coefficient R, and RMSE are also shown.

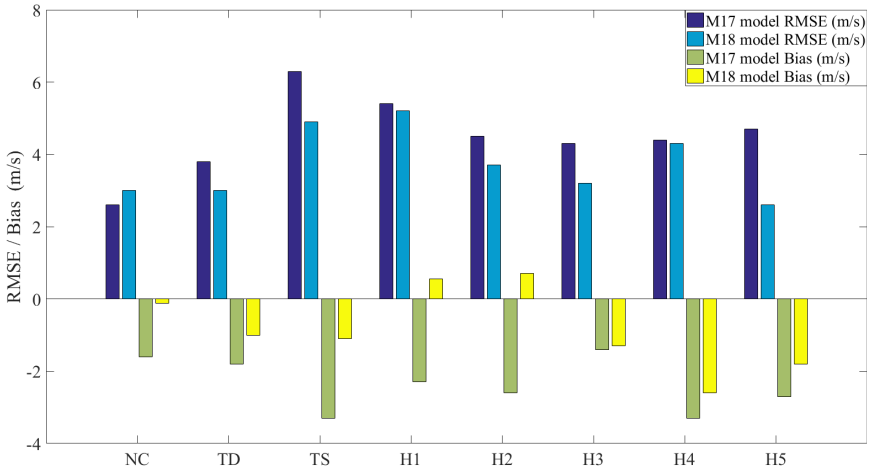


Fig. 8. RMSE and bias of the TC intensity estimated by $M17$ and $M18$ models.

TABLE XI
CLASSIFICATION REPORT OF THE BINARY CNN CLASSIFICATION MODEL $M18$

Category	P	R	F1	Total
NC	0.69	0.92	0.79	35
TD	0.92	0.91	0.91	386
TS	0.93	0.93	0.93	552
H1	0.85	0.83	0.84	178
H2	0.73	0.78	0.76	85
H3	0.84	0.74	0.79	62
H4	0.83	0.86	0.85	36
H5	1	1	1	4
Avg.	0.89	0.89	0.89	1338

variability among TCs over the Northwest Pacific [15], [42], which makes it more challenging to estimate their intensities accurately. The accuracy of our model over the Northwest Pacific is comparable to the accuracy of the multiclassification CNN model proposed by Pradhan *et al.* [18] over the global ocean.

The performance of CNN models with the same structure may have some differences due to different training data. Therefore, we used the 6.690 four-channel H-8 images of TCs collected to retrain Chen's and Tian's model [15], [19]. The results (Table XII) show that our binary CNN classification model performs better than the other two CNN models if the same satellite data were used, which further demonstrates our model's robustness.

3) Noise Resistance Ability of the CNN Model: As shown in Fig. 9, satellite images sometimes include noise, i.e., some distinct black spots on the Channel 7 brightness temperature images. To determine the extent to which different levels of

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TABLE XII
COMPARISON OF THE PERFORMANCES OF OUR MODEL AND OTHER TECHNIQUES IN ESTIMATING THE MWS OF TCS

Literature	Technique	Region	RMSE (m/s)
Kossin et al. [47]	Linear regression	Global ocean	6.79
Ritchie et al. [48]	DAVT	North Atlantic	6.53
Fetanat et al. [49]	K-nearest-neighbors algorithm	Global ocean	6.53
Pradhan et al. [18]	CNN classification	Global ocean	5.24
Chen et al. [15]	CNN regression (without smoothing)	Global ocean	5.44
		Northwest Pacific	6.30
		Northwest Pacific (using our data)	5.80
Tian et al. [19]	CNN classification and regression	Global ocean	4.58
		Northwest Pacific (using our data)	4.89
Our model	**Binary CNN classification**	**Northwest Pacific**	**4.62**

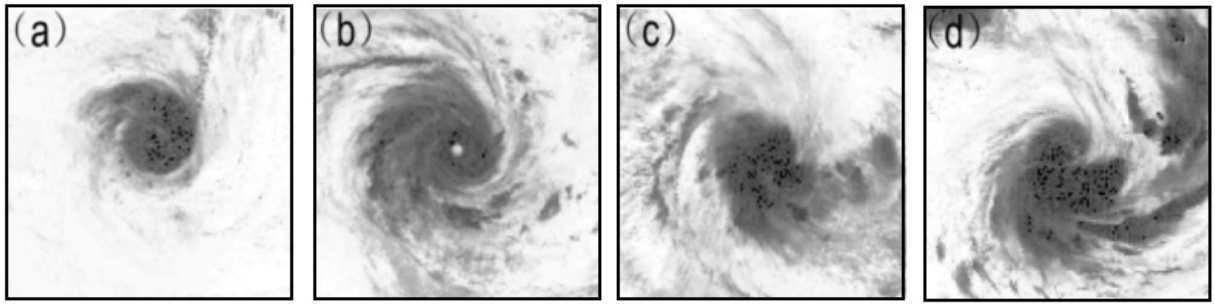


Fig. 9. Brightness temperature images from Channel 7 with noise (black spots). (a) TC Halola (18:00 UTC on July 21, 2015), (b) TC Soudelor (15:00 UTC on August 4, 2015), (c) TC Tokage (12:00 UTC on November 24, 2016), and (d) TC Soulik (18:00 UTC on August 16, 2018).

noise affect our model's performance, we carried out some antinoise experiments by attacking the optimal network with several adversarial attack methods [43], [44]. The adversarial attack means adding artificial noise to the input image and tricking a neural network into obtaining incorrect classification results. For example, for a TS, if we attack our model using an adversarial attack method (or noise adding method), the model may classify it as an H3 TC. In this way, we aim to test the stability or antinoise ability of our model.

Among all the models in Table V, the $M16$ model generates the best classification results. In this study, we applied six gradient-based adversarial attack methods to test the $M16$ model's stability [43]–[45]. These attack methods include Random Projected Gradient Descent (Random PGD), Fast Gradient Sign Method (FGSM), Linfinity Basic Iterative Attack (LBIA), Basic Iterative Method (BIM), Saliency Map Attack (SMA), and Linfinity Basic Fast Gradient Sign Attack (LBFGSA) to test the stability of this model. When the loss of a neural network increases to a specific value by artificially adding some noises with an adversarial attack method, the image is misclassified. Here, we consider a case where TS is misclassified as H3. Fig. 10 shows examples of the original images and the noise added to the images using different methods. One can see that the FGSM method [Fig. 10(b)] fails to misclassify our model, suggesting that the model is reasonably stable at this noise level. The other five methods succeeded in misclassifying the CNN-I model to some extent, but apparent anomalies can be observed in the images after adding noise by four methods (i.e., the LBIA, BIM, SMA, and LBFGSA methods). One can see that the accuracy of our model only decreases when we add very high-level noise. On the other hand, images with substantial noise can be easily identified and removed from the input data, implying that it is difficult for the four methods to influence our model performance in practice. Among the six methods, Random PGD is the only one that may misclassify the TC category effectively. To know how vulnerable our model is to the noise added by the Random PGD method, we analyzed its capability in classifying a TS with different attacking powers. Here, we use the noise-to-signal (NS) ratio, i.e., the absolute value of the ratio between the noise added to the signal and the original signal at each pixel of the image, to represent the adversarial attack power. Fig. 11 shows the variation of the confidence with which TS is misclassified as the $H3$ category with the NS ratio. As the attack power increases, the confidence of misclassification increases as well. However, even under the lowest attack power, we can still observe that the image with added noise is rougher and fuzzier than the original image in Fig. 10. This result means that our model can defend against this adversarial attack method when the NS ratio is below 2%.

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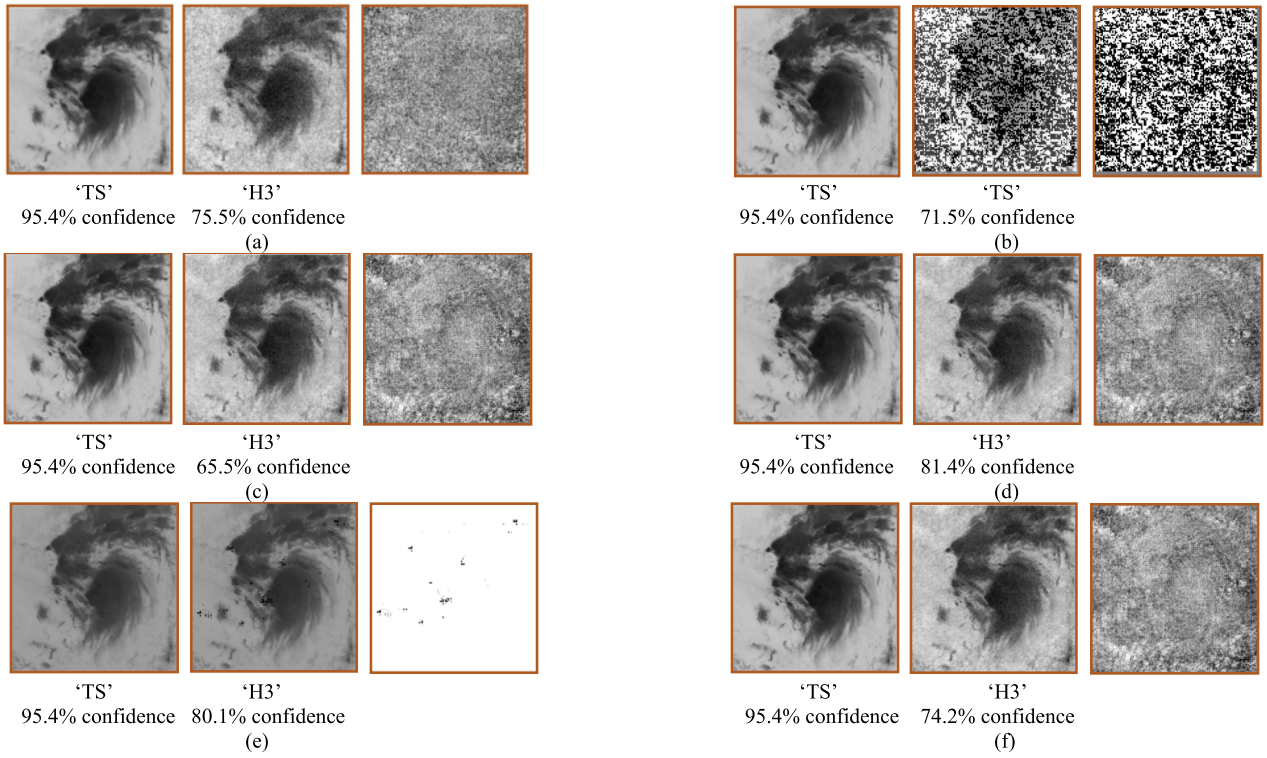


Fig. 10. Adversarial attack experiment results using various attacking methods. (a) Random PGD, (b) FGSM, (c) LBIA, (d) BIM, (e) SMA, and (f) LBFGSA. For each figure, the left column is the original image from Channel 15, the middle column is the image after adding noise, and the right column shows the noise added to the original image.

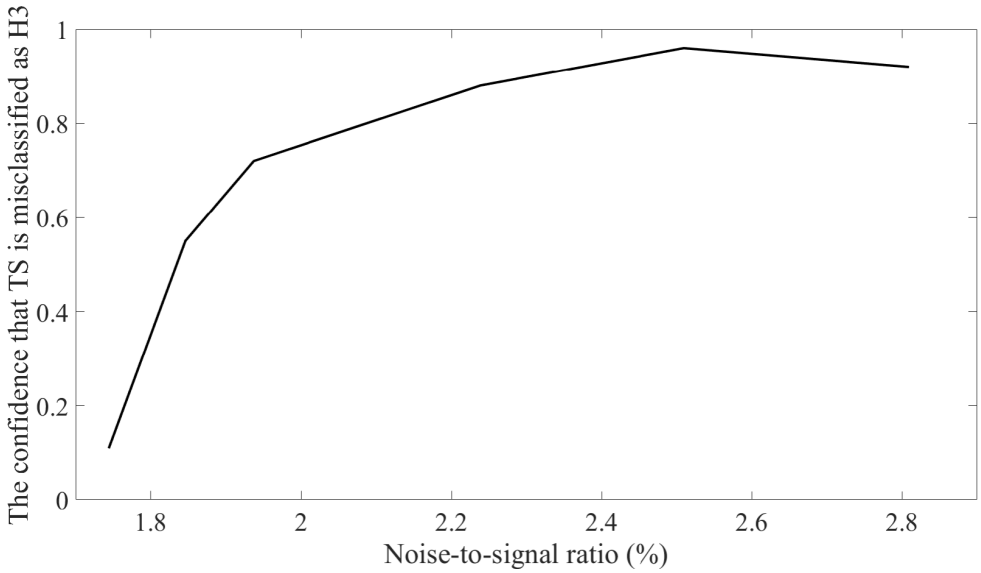


Fig. 11. Variation of the confidence with which TS is misclassified as $H3$ with NS ratio. The noise was added to the satellite images using the Random PGD method.

The traditional methods do not consider noise, which may affect results, so we cannot measure noise's influence on conventional methods. The above experiments prove that our CNN model can be unaffected by noise to a particular range.

V. Conclusion

TCs are one of the most severe disasters. Estimating their intensity is an essential step for forecasters and emergency responders to make disaster warnings. This article developed utterly objective CNN-based models to estimate TC intensity from H-8 geostationary satellite IR imagery, which can provide a reliable basis for TC prediction and early warning systems.

Results show that the selection of satellite data channels has a significant impact on the TC intensity estimation models performance. Network parameters play an essential role in CNN models. Using four-channel (Channels 7, 8, 13, and 15)

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IR imagery, we found that the CNN model $M16$, which includes one input layer, one channel attention layer, one spatial attention layer, four convolutional layers, four pooling layers, two FC layers, and one output layer, has the best results among the multiclassification models. The model has generated a fairly good accuracy (86.0%), low root mean square error (RMSE, 5.17 m/s), and low bias (−2.15 m/s).

Several other factors affect the accuracy of the CNN-based TC intensity model, such as the TC structure, and the imbalance of TC category samples. Our study shows that the TC eye, size, shape, and brightness temperature anomaly are the main factors possibly influencing the model performance.

For TC categories with much smaller training data set, the multiclassification model does not perform very well due to the TC categories' unbalanced distributions. In this study, for the first time, we introduced the FL function in the network to increase the weight of the TC category with fewer data in loss, and we adopted eight binary classification networks to reduce the side-effect of the unbalanced training data. The new binary classification model can better learn the features of these categories with fewer samples. The number of exact-hits for NC, TD, TS, $H1$, $H2$, $H3$, and $H4$ is increased. In particular, the precision of $H3$ classification is significantly improved by 9.1%. Compared with the multiclassification model, the binary classification model $M18$ generates a higher accuracy (88.9%), and much lower RMSE (4.62 m/s) and bias (−0.76 m/s) in TC intensity estimation. The results are comparable to or even better than some of the state-of-the-art TC intensity estimation techniques.

In this work, we mainly used the CMA Best Track data set to label satellite images and train the TC intensity estimation model. As is known, any biases in the Best Track data set may introduce biases into the model. The difference in the Best Track data sets provided by different operational agencies such as the Japan Meteorological Agency (JMA) and Joint Typhoon Warning Center (JTWC) may also affect the model performance. Assuming that these biases are mainly caused by the differences in TC center positions provided by different agencies, we then used the JMA and JTWC determined TC center positions to re-extract satellite images in the testing data set for TCs between 2015 and 2018. The comparison between estimated TC intensity and the corresponding Best Track data shows that our CNN model also has very high accuracy with the RMSE of 4.72 and 4.70 m/s, respectively, quite close to that estimated based on CMA Best Track data (4.62 m/s). Note that the mean difference between the CMA TC center and JMA/JTWC TC center is 11.2 km/10.3 km. The result indicates that such a limited TC positioning error does not affect the performance of our CNN model greatly. For further development of the model, it is necessary to collect more satellite data such as HURSAT data to explore the TC characteristics and improve the CNN-based TC intensity estimation model [50].

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Chong Wang received the B.S. degree in marine science from Hohai University, Nanjing, China, in 2018, where he is pursuing the master's degree in physical oceanography.

His research interests include tropical cyclone monitoring and forecasting using remote sensing and artificial intelligence techniques.



Gang Zheng (Senior Member, IEEE) received the B.Eng. degree in electronic information engineering from Zhejiang University, Hangzhou, China, in 2003, and the M.S. and Ph.D. degrees in radio physics from the University of Electronic Science and Technology of China, Chengdu, China, in 2006 and 2010, respectively.

From 2010 to 2013, he was an Assistant Researcher with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, where he was an Associate Researcher, from 2013 to 2020, and has been a Researcher, since 2020. His research interests include ocean microwave remote sensing, artificial intelligence applications, image processing, and electromagnetic numerical modeling.

Dr. Zheng is an Editorial Board Member of the Ocean Section of *Remote Sensing* and a Topic Editor of the *Big Earth Data*. From 2018 to 2020, he also served as the Guest Editor for *Remote Sensing*, Special Issues on AI-based Remote Sensing Oceanography, Synergy of Remote Sensing and Modelling Techniques for Ocean Studies, and Tropical Cyclones Remote Sensing and Data Assimilation.

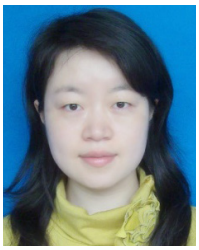


Xiaofeng Li (Fellow, IEEE) received the B.S. degree in optical engineering from Zhejiang University, Hangzhou, China, in 1985, the M.S. degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 1992, and the Ph.D. degree in physical oceanography from North Carolina State University, Raleigh, NC, USA, in 1997.

He was with the National Environmental Satellite, Data, and Information Service, National Ocean and Atmospheric Administration, College Park, MD, USA, from 1997 to 2019. His research interests include synthetic aperture radar (SAR) applications in oceanography and marine meteorology, artificial intelligence oceanography, big data, and satellite image processing.

Dr. Li is an Editorial Board Member of the *International Journal of Digital Earth*, *Big Earth Data*, and *Journal of Oceanology and Limnology*. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the *International Journal of Remote Sensing*. He is also the Executive Editor-in-Chief of the *Journal of Remote Sensing* (a Science Partner Journal).

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Qing Xu (Member, IEEE) received the B.S. degree in marine science and the Ph.D. degree in physical oceanography from the Ocean University of China, Qingdao, China, in 2002 and 2007, respectively.

In 2006, she joined the Department of Atmospheric and Oceanic Science, University of Maryland, College Park, MD, USA, as a Visiting Scholar. From 2007 to 2009, she was a Post-Doctoral Fellow with the Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Hong Kong. From 2009 to 2016, she was an Associate Professor with the Department of Physical Oceanography, Hohai University, Nanjing, China, where she has been a Full Professor since 2016. From 2015 to 2016, she was a Visiting Scholar with the Center for Coastal Physical Oceanography, Old Dominion University, Norfolk, VA, USA. She is with the College of Information Science and Engineering, Ocean University of China. Her research interests include the application of satellite remote sensing and artificial intelligence in oceanography and meteorology.

Dr. Xu served as the Guest Editor of a Special Issue on Satellite Remote Sensing of Weather, Water and Climate Couplings and Phenomena for the *Remote Sensing* (2018–2019).



Bin Liu (Member, IEEE) was born in Hunan, China, in 1985. He received the B.Sc. degree in information engineering and the M.Sc. and Ph.D. degrees in signal and information processing from Shanghai Jiao Tong University, Shanghai, China, in 2007, 2009, and 2015, respectively.

From 2012 to 2013, he was a Visiting Ph.D. Student with the Department of Signal and Image Processing, Télécom ParisTech, Paris, France. From 2015 to 2018, he was a Research Assistant Professor with the Shanghai Key Laboratory of Intelligent Sensing and Recognition, Shanghai Jiao Tong University. From 2018 to 2020, he was a Post-Doctoral Fellow with the College of Marine Sciences, Shanghai Ocean University, Shanghai. Since 2020, he has been a Full-Time Teacher with the College of Marine Sciences, Shanghai Ocean University. His research interests include Artificial Intelligence Oceanography, synthetic aperture radar (SAR) and polarimetric SAR image processing and understanding, and image segmentation and classification.

Dr. Liu is an Active Reviewer for several journals, including the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jun A. Zhang (Senior Member, IEEE) received the B.S. degree in naval architecture and ocean engineering from the Dalian University of Technology, Dalian, China, in 2000, and the Ph.D. degree in applied marine physics from the Rosenstiel School of Marine and Atmospheric Science, University of Miami, Miami, FL, USA, in 2007.

He is a Scientist with the NOAA/Atlantic Oceanographic and Meteorological Laboratory/Hurricane Research Division and University of Miami. His research interests include boundary layer structure and dynamics, turbulence, air–sea interaction, remote sensing, and physical parameterizations.

Dr. Zhang was a Post-Doctoral Fellow of the National Research Council. He received the CIMAS Gold Medal Award in recognition of his role in improving the operational Hurricane Weather Research and Forecasting (HWRF) Model, the American Meteorological Society's Banner I. Miller Award, the NASA Award for Hurricane and Severe Storm Sentinel Group Achievement, and the Aviation Laureate Award. He has been serving as the Chief Editor for the *Dynamics of Atmospheres and Oceans*, since 2019.

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